Disclosure Regulation, Intangible Capital, and the Disappearance of Public Firms[†]

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Abstract

Since the mid-1990s, the number of U.S. listed firms has halved while their performance has become harder to predict. We develop an analytically tractable general equilibrium model where firms choose between public and private markets and select disclosure levels. In the estimated model, a rising intangible share reduces public-market participation, productivity, and welfare by weakening knowledge diffusion. Welfare exhibits an inverted U shape in mandated exposure: insufficient disclosure forgoes knowledge diffusion, while excessive disclosure drives firms private. We characterize the optimal disclosure policy and find that the data-fitted mandates lie significantly above it: justifying them requires the wedge in the external benefit worth about 13 percent of benchmark welfare in 1992–1996 and 29 percent in 2012–2016 under the calibrated spillover elasticity. We show that this doubled wedge in recent years is due to the economy becoming intangible-intensive.

Keywords: Disclosure regulation, public-market participation, intangible capital, information frictions.

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1 Introduction

> *"As a smaller private company, Google kept business information closely held, and we believe this helped us against competitors... As a public company, we will of course provide you with all information required by law... But we will not unnecessarily disclose all of our strengths, strategies and intentions."*
>
> — Larry Page and Sergey Brin, *A letter from Google founders to the shareholders, 2004*

The U.S. public equity market has undergone a dramatic transformation: listed firms declined from around 8,090 in 1996 to 4,102 in 2012, a 49% contraction, even as the overall firm population continued to grow. Concurrently, we document that analyst forecast errors for public firms rose more than three times indicating deterioration in market transparency. What forces simultaneously reduce public listings and increase opacity among remaining firms? What are the aggregate welfare and productivity consequences?

This paper makes two contributions. First, we develop a closed-form framework that captures how firms' incentives to protect proprietary information and investors' preferences for transparency jointly shape the allocation of firms across listed and non-listed markets. Using this framework, we analyze the forces driving the observed patterns and their macroeconomic implications. Notably, our formulation yields an explicit analytical expression for the measure of listed firms, enabling general-equilibrium comparative statics with respect to key structural parameters. We show that, in general equilibrium, the number of listed firms strictly decreases in both intangible intensity and regulatory stringency.

Second, based on our estimated model, we quantitatively evaluate the effects of disclosure regulation on the number of listed firms and welfare, and derive the optimal policy. Stricter regulation improves investor information but drives firms toward non-listed markets, reducing the pool of transparent firms and knowledge spillovers.[1] In the estimated model, welfare exhibits an inverted-U relationship with mandated exposure: knowledge diffusion supports

[1] This is one of the core issues the SEC is concerned about. For example, in a February 2017 speech, SEC Commissioner Kara Stein questioned whether additional disclosure requirements should be extended to private-market firms: "We also need to understand why more companies are staying private for longer periods of time. Should we apply enhanced disclosure laws to these private companies? Or perhaps they require a unique set of rules." The full speech is available on the SEC's website. The concern is empirically grounded: Dambra et al. (2015) study Title I of the JOBS Act, which relaxed disclosure requirements for emerging growth companies during the IPO process. They find a 25% increase in IPO volume, driven largely by firms with high disclosure costs, such as biotechnology and pharmaceutical firms. Aghamolla and Thakor (2022) exploit a shock to disclosure requirements in the biopharmaceutical industry to show that increased mandatory disclosure requirements for private firms significantly increase their propensity to go public. Finally, Abuzov et al. (2025) document that court rulings in late 2002, which prevented public investors from keeping fund information confidential, prompted many top VCs to exclude such investors from their funds.

an interior optimum. According to our model, the observed regulatory requirements are substantially above the welfare-maximizing level. We characterize the distance between the optimal and the observed levels in the welfare criterion of the model: the wedge outside the model's welfare criterion, such as fraud deterrence or agency discipline, would need to be worth about 13 percent of benchmark welfare to justify the fitted 1992–1996 requirement and about 29 percent for the 2012–2016 requirement. Our decomposition reveals that rising intangible intensity, together with the spillover elasticity tied to it, has reduced welfare by 1.91 percent annually, while the estimated rise in regulatory exposure has reduced welfare by 0.68 percent annually, along with a 0.55 percent annual productivity decline. Critically, we show that the welfare consequences of disclosure policy have become substantially more pronounced in the intangible-intensive economy.

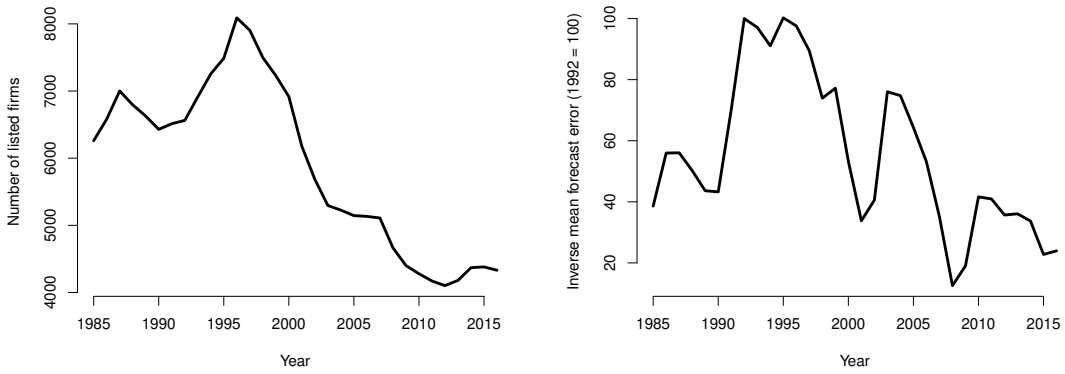
Our model captures a fundamental tension in disclosure. Firms with intangible capital face spillover costs: disclosed knowledge, whether R&D, organizational practices, or business strategies, diffuses to competitors, reducing the rents the disclosing firm can exclusively capture.[2] Financiers, conversely, prefer to invest in transparent firms where information precision is higher. We model this through transparency submarkets, where funding supply increases with transparency (as financiers allocate more capital to better-understood firms) while funding demand decreases with required disclosure (as spillovers erode competitive advantage). Funding in both markets is supplied by specialized financiers who hold claims only in their own market segment, so firm-level risk is priced even though the economy has no aggregate risk. In equilibrium, ex-ante identical firms endogenously sort into a continuous distribution of transparency levels, with the density characterized in closed form. Critically, rising intangible intensity amplifies firms' disclosure costs, shifting the equilibrium toward greater opacity and more private firms, consistent with the following observed patterns.

Figure 1 plots the time series for the number of listed firms in the U.S. (panel 1a) and the inverse forecast error (panel 1b), which proxies the overall transparency in the stock market.[3] According to panel (a), there has been a gradually rising trend in the number of listed firms until the mid-1990s. Then, after the peak in the mid-1990s, the number of listed firms steeply declined to almost half the level at the peak year: 8,090 listed firms in 1996

[2]We refer to those components of intangible capital whose property rights are not well protected by specific legal institutions and thus not necessarily patentable or patented yet. For example, software, research ideas, early stages innovation and R&D, and also certain novel business methods and organizational innovations, branding and marketing strategies, employee training, information such as some formulas, customer lists, and processes; more in general, firms' strategies and intentions that a public firm cannot selectively disclose (e.g., Regulation Fair Disclosure of 2000).

[3]The detailed measurement is available in Section 4, which aims at explicitly connecting intangible capital and different transparency proxies, controlling for confounders. Additional analyses on industry-level time series are available in Appendix B.

Figure 1: Time series of aggregate variables.



(a) Number of listed firms

(b) Inverse forecast error

Notes: The figure shows the number of listed firms and inverse analyst forecast error in the United States. Panel (b) plots the inverse of the mean absolute earnings surprise, indexed to 100 in 1992, constructed as described in Section 4. Data come from the World Development Indicators (panel a) and Compustat and I/B/E/S (panel b).

reduced to 4,102 listed firms in 2012. Listed firms have been declining not only in absolute number, but also as a share of all firms in the U.S. In Appendix B, we show that the decline is substantially larger in industries like information and services, which are more intangible intensive. This is consistent with recent empirical evidence showing that firms with higher intangibility have a significantly lower probability of being publicly listed, controlling for other characteristics (Celentano and Rempel, 2026). In Appendix C, we relate industry-level listing, entry, and exit rates to intangible density and lagged opacity.

Panel (b) plots the time series of the inverse forecast error. The overall patterns of the series closely mimic those in the number of listed firms: the inverse forecast error has increased until the mid-1990s and decreased after the peak in 1995.[4] We show that the decline in public-market participation and overall greater opacity in financial markets substantially reduce productivity through the weakened technological diffusion across firms, which is consistent with the recent slowdown in U.S. productivity growth and business dynamism (Gordon, 2017; Bloom et al., 2020; Akcigit and Ates, 2023). According to our baseline model, it results in a substantial welfare loss in the economy.

In Section 4, we empirically examine the relationship at the firm level. Intangible intensity is significantly associated with larger analyst forecast errors, controlling for observable

[4]Recessions and especially the Great Recession represent a big shock to earnings surprises. In order to take that into account, we also measure the inverse forecast error by excluding recession periods as measured by the NBER, and we still find that the time series has been declining.

characteristics and fixed effects. The association is concentrated in internally generated intangible capital, such as knowledge capital from R&D and organizational capital. Acquired intangibles recognized on the balance sheet predict better forecastability. The aggregate deterioration in forecastability is robust over the asset-weighted firm-level average.

Disclosure regulation fundamentally reshapes market equilibrium through two channels. Stricter requirements increase mandated transparency, directly improving information quality for listed firms. However, this intensifies competitive spillovers, inducing more firms to exit public markets. Financiers' information gains are capitalized into equilibrium funding terms, so the welfare value of a mandate comes from knowledge diffusion: welfare peaks at a moderate exposure index and declines steeply beyond it. Productivity is also hump-shaped in the exposure index. This trade-off has intensified substantially in the recent intangible-intensive economy: the same deviation from the optimal policy now generates larger welfare losses than in the 1990s.

Related literature Three strands of the literature are closely related to this paper. The first is the literature that studies the rising importance of intangible capital for the macroeconomy. Early contributions, including Atkeson and Kehoe (2005), McGrattan and Prescott (2010), Eisfeldt and Papanikolaou (2014), and McGrattan (2020), highlight the importance of intangible capital as a key input factor for production. Recent research links the increase in intangible capital to several macroeconomic shifts: the slowdown of productivity growth and the rise of market power (De Ridder, 2024); the rise in firm size and concentration (Chiavari and Goraya, 2025); rising cash holdings (Falato et al., 2022); lower profit taxes under international tax competition (Quadrini and Ríos-Rull, 2024); the corporate savings glut (Li, 2025).

We contribute to this literature by analyzing a novel macroeconomic implication of the rising share of intangible capital. Intangible capital has become an important source of competitiveness, leading firms to invest heavily in research and development (R&D) and organizational practices. However, many forms of intangible capital have strong spillover effects, benefiting other firms as well as the originating firm (Bloom et al., 2013; Haskel and Westlake, 2018; Crouzet et al., 2022). The rising importance of intangible capital has therefore increased firms' incentives to stay opaque in disclosure. Our model quantifies how this shift affects aggregate outcomes and welfare.

Within this literature, our paper is closely related to Celentano and Rempel (2026) and Ward (2025). Celentano and Rempel (2026) find that the rising share of intangible capital amplifies CEOs' private information relative to outside investors, increasing the cost of optimal truth-telling contracts and lowering the propensity to go public. Ward

(2025) shows that intangibles exacerbate internal agency conflicts with specialists, generating underinvestment and rising dispersion in specialists' compensation. We abstract from these agency frictions and focus instead on a complementary channel: public disclosure to investors can also make competitively sensitive information accessible to rivals, forcing a trade-off between funding access and *competitive spillovers* (Bhattacharya and Ritter, 1983). Our closed-form general equilibrium framework enables normative analysis, characterizing the welfare effects of disclosure policy and quantifying welfare costs as intangible intensity rises.

Celentano and Rempel (2026) also study listing choices in general equilibrium with endogenous private-equity funding. Compared to their model, the distinctive feature of ours is the disclosure-induced knowledge spillover across firms, which lowers the disclosing firm's rents but raises other firms' productivity. This spillover gives listing its social value, and it links the decline in listings to aggregate productivity.

The second literature strand is the one that studies the incentive for information disclosure and its real impact. One of the seminal papers in the literature is Hirshleifer (1971), which studies how information disclosure can be incentivized through pecuniary motivation, which is closely related to the firms' incentive for transparent disclosure in our model. Subsequent theoretical and quantitative work examines disclosure regulation (Admati and Pfleiderer, 2000; Terry et al., 2023), firms' disclosure choices (Boot and Thakor, 2001; Bertomeu et al., 2022), and how disclosure interacts with financing and ownership structure (Bhattacharya and Ritter, 1983; Bhattacharya and Chiesa, 1995; Chemmanur and Fulghieri, 1999; Boot et al., 2006, 2008; Chemmanur and He, 2011; Ferreira et al., 2014). Terry et al. (2023) quantify how disclosure regulation affects earnings misreporting, intangible investment, growth, and welfare in general equilibrium. We focus instead on the competitive spillovers of disclosure and on endogenous public-market participation. We contribute to this literature by showing that the rising share of intangible capital interacts with disclosure regulation to generate time-varying incentives for transparency. Embedding these incentives in general equilibrium allows us to quantify their macroeconomic implications on output and productivity and to evaluate the welfare consequences of disclosure mandates.

The third literature is about the disappearance of listed firms. Different explanations have been put forward to shed light on this issue. For example, Gao et al. (2013) point to the growing advantage for small firms of selling out to larger firms rather than going public; Doidge et al. (2017) find evidence consistent with a decrease in the net benefits of going public for U.S. firms; Ewens and Farre-Mensa (2020) argue that the deregulation of securities laws (National Securities Markets Improvement Act of 1996) improved the private equity market, which reduced the incentives for firms to go public; Davydiuk et al. (2020) point to both a significant reduction in private firms' cost of capital and an increase in the

costs of being public and undergoing an IPO.

This paper provides a complementary explanation. We argue that the rise of intangible capital, especially the components of intangible capital that could benefit competitors as well as the owner firm, has increased the cost of disclosing information and made staying private more attractive, which is exacerbated by stricter disclosure requirements. Our framework builds on the proprietary-information trade-off emphasized by Doidge et al. (2018) and adds endogenous listing, knowledge diffusion across firms, and the welfare effects of disclosure mandates. The mechanism is also consistent with the sectoral patterns documented in Appendix B and with evidence linking greater firm intangibility to a lower probability of being publicly listed (Celentano and Rempel, 2026).

Roadmap The rest of this paper proceeds as follows. Section 2 presents an analytically tractable general equilibrium model. Section 3 analyzes the equilibrium. Section 4 provides empirical evidence supportive of the theory's predictions. Section 5 presents a quantitative analysis and analyzes the optimal disclosure policy. Section 6 concludes.

2 Model

We develop a static general equilibrium model, where a firm's decision of where to operate between the public and private financial market, the specialized financiers' local funding choices, and the cross-sectional distribution of disclosure transparency are all analytically characterized.[5]

2.1 Segmented financiers

Public-market funding is supplied by specialized financiers. The funding side is indexed by cells (q, j): for every potential disclosure submarket $q \in [0, 1]$, a unit mass of ex-ante identical financiers $j \in [0, 1]$ is assigned to that submarket, and a mass $1/\nu_N < 1$ of financiers serves the private market, reflecting thinner participation in private-firm evaluation. Only cells with $q < 1-\overline{q}$ are active where $\overline{q}$ is the mandated transparency level; an operating cell should abide by the mandate. We assume that specialized financier (q, j) holds only the claim package of its own cell. This limited-participation structure captures the specialization in narrow groups of firms among financial analysts, which is motivated by investor recognition (Merton, 1987)

[5]The model is static and intended to capture a long-run equilibrium. The static setup delivers substantial tractability, yielding closed-form solutions characterized in Section 3. The tangible capital rental rate is exogenous, as in a small open economy. This simplification allows us to focus on the disclosure and listing decisions while maintaining analytical tractability.

and information-based portfolio concentration (Van Nieuwerburgh and Veldkamp, 2010). Thus, the information structure closely follows that of the I/B/E/S earnings forecast data our model quantification is based on. We assume the symmetric choice $x_j(q) = x(q)$ for all j, so aggregate funding to submarket q is $\int_0^1 x_j(q)\,dj = x(q)$. Therefore, the per-financier choice $x(q)$ is the funding density for that cell.[6]

Each financier has CARA utility $u(C) = -e^{-\Lambda C}$, where $\Lambda > 0$, and access to a risk-free outside asset whose excess return is normalized to zero. The excess return available to financier j in listed submarket q is

$$\widetilde{r}_j(q) \sim N\left(\overline{r}(q), H(q;\overline{q})\right), \qquad \overline{r}(q) = \frac{\pi(q)}{P(q)}, \tag{1}$$

where $\pi(q)$ and $P(q)$ are the expected profit and price of a firm with transparency q. The realized return on financier (q, j)'s nondiversifiable package of claims equals its mean $\overline{r}(q)$ plus a mean-zero *idiosyncratic payoff shock*: the shocks are cell-specific, independent across financiers and submarkets, and carry no aggregate component. Disclosure lowers the residual payoff uncertainty an outside financier faces: the variance $H(q;\overline{q})$ declines with transparency because reported information resolves, before returns realize, part of what the financier does not know about the firm, which links H to its empirical counterpart, the dispersion of analyst forecasts (Section 4). Financier j's terminal wealth is $C_j(q) = a_j(q) + x_j(q)\widetilde{r}_j(q)$, where $a_j(q)$ includes the payoff from the outside asset. Under CARA-normality, choosing local funding is equivalent to maximizing the money-metric certainty equivalent

$$CE_j(q, x_j) = a_j(q) + x_j\overline{r}(q) - \frac{\Lambda}{2}x_j^2 H(q;\overline{q}), \quad x_j^*(q) = \max_{x_j}\ CE_j(q, x_j). \tag{2}$$

We assume each financier's wealth capacity is sufficiently large that the upper bound on $x_j(q)$ does not bind.

We assume that the information noise $H = H(q;\overline{q})$ strictly decreases in the level of transparency q and the mandated transparency $\overline{q}$, which is a policy parameter to be estimated. Throughout the paper, H is also referred to as the financiers' forecast-error variance. For the analytical tractability, we assume the following parametric form of H:

$$H(q;\overline{q}) = \frac{1}{\xi + \psi(\overline{q} + q)} \tag{3}$$

where ξ is the minimum information level that is *commonly* available about both listed

[6]The measure of active public cells falls with the mandate: part of the listing response to $\overline{q}$ reflects the elimination of low-transparency market segments and the exit of their specialized financiers.

and non-listed firms, and ψ is the marginal contribution of transparency to the financiers' information about the listed firm.[7]

The private-market local return is analogously

$$\widetilde{r}_j^N \sim N\left(\overline{r}^N, H(0;0)\right), \qquad \overline{r}^N = \frac{\pi^N}{P^N}, \tag{4}$$

where π^N and P^N are the profit and price of a non-listed firm. As non-listed firms are assumed not to disclose any information publicly, their financiers face $H(0;0) = 1/\xi$. Their net certainty equivalent and corresponding optimal funding are

$$CE_j^N(x_j^N) := x_j^N \overline{r}^N - \frac{\Lambda}{2}(x_j^N)^2 H(0;0), \quad x_j^{N*} := \max_{x_j^N} \; CE_j^N(x_j^N) \tag{5}$$

where the policy-invariant outside-wealth term is omitted.

It is important to note that there is no aggregate risk in this economy: an exact law of large numbers applies to the independent cell-specific payoff shocks, which wash out in the aggregate, so aggregate output and consumption are deterministic. Crucially, financiers cannot share those shocks: each bears the payoff uncertainty of its own cell when choosing $x_j(q)$ or x_j^N. Thus local risk is priced even though the aggregate is riskless.

2.2 Technology

A measure one of the ex-ante homogeneous firms produces output using two inputs: tangible capital (k_T) and intangible capital (k_I). In this economy, there are two types of production technologies. One is *listed* firms' production technology, and the other is *non-listed* firms' production technology. Before production, a firm chooses 1) which market to operate in between the listed and non-listed market and 2) the level of transparency q conditional on the choice of the listed market. We restrict the possible range of q to $[0, 1 - \overline{q}]$. The policy variable $\overline{q} \in [0, 1]$ is a normalized mandated-disclosure index, and q is a normalized voluntary disclosure index. Hence $y_i := \overline{q} + q_i \in [0, 1]$ measures the total disclosure-induced exposure of firm-specific rents; neither q_i nor $\overline{q}$ is a physical share or unit of intangible capital.

[7]The parameter ψ may itself depend on intangible intensity θ: as intangibles become more important, the same disclosure conveys less information to investors. We do not impose structure on this relationship, instead identifying ψ directly from firm-level data. ξ captures baseline opacity, which may partly reflect strategic concealment as in Bertomeu et al. (2022). Our model treats ξ as exogenous, though the estimates incorporate any such endogenous channel in reduced form.

2.2.1 Production function of listed firms

A listed firm i operates using the following production function:

$$f^L(k_i^T, k_i^I, q_i; \overline{q}, \Phi^{ex}) = (k_i^T)^\alpha (k_i^I(1 - \overline{q} - q_i))^\theta (\Phi^{ex})^\gamma, \tag{6}$$

where Φ^{ex} is the stock of shared knowledge generated by other public firms, γ is the scale parameter for that externality, and α and θ are the tangible and intangible input elasticities, respectively. We assume $\alpha + \theta + \gamma \leq 1$.

The factor $1 - y_i$ is retained appropriability: it is the fraction of the services or rents generated by $k_{I,i}$ that the firm can capture after its strategy, know-how, and other competitively sensitive information become exposed. Disclosure therefore does not destroy $k_{I,i}$ or mechanically transfer the asset. It enables competitors, customers, and suppliers to respond, which dissipates firm-specific rents while also adding to a common knowledge pool. The linear factor is a reduced-form normalization; in particular, a value of y_i near one means high rent exposure, not the disappearance of an equal percentage of the firm's intangible stock.[8]

We assume the exposed knowledge contributions are perfectly substitutable across firms. Therefore, the shared knowledge is aggregated in the following additive form:[9]

$$\Phi^{ex} = \int_0^1 1_{\{i \in \text{Listed}\}} \times k_{I,i} \left(\underbrace{\overline{q}}_{\text{Disclosure mandated by the policy maker}} + \underbrace{q_i}_{\text{Voluntary disclosure}} \right) di. \tag{7}$$

This equation maps the normalized exposure index into shared knowledge; it is not a conservation identity for intangible capital.

The expected profit of listed firm i with voluntary transparency q_i, defined as the mean of the payoff around which the idiosyncratic shock realizes, is obtained after paying out the

[8] At the limiting normalization $y_i = 1$, all firm-specific rents from the exposed intangible are competed away. Legally protected ownership may remain intact. The quantitative intangible share is estimated from intangible-related expenditures, so the model does not require a literal mapping from y_i into units of an intangible stock.

[9] More generally, one could allow for a finite elasticity of substitution $\lambda < \infty$ in the following setup:

$$\Phi^{ex} = \int_0^1 1_{\{i \in \text{Listed}\}} \times k_{I,i} \left(\overline{q}^{\frac{\lambda-1}{\lambda}} + q_i^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}} di.$$

In our baseline setup, we take $\lambda \to \infty$. Allowing for a finite λ does not affect the theoretical predictions, however, because the equilibrium firm distribution is isolated from price and externality effects. Propositions 3 and 4, along with Corollary 3, make this independence explicit.

operating costs $rk_i^T + pk_i^I$ from the revenue:

$$\pi(q_i; \overline{q}, \Phi^{ex}) := \max_{k_i^T, k_i^I} \quad (k_i^T)^\alpha (k_i^I(1 - \overline{q} - q_i))^\theta (\Phi^{ex})^\gamma - rk_i^T - pk_i^I, \tag{8}$$

where r is the capital rental rate, and p is the $R\&D$ cost per unit of intangible capital. For the notational brevity, we assume the levels of r and p already include the depreciation (amortization) rates.

2.2.2 Production function of non-listed (private) firms

If a firm is private, it does not publicly disclose the intangible capital. The production function of a non-listed firm i is as follows:

$$f^N(k_i^T, k_i^I; \Phi^{ex}) = (k_i^T)^\alpha (k_i^I)^\theta (\Phi^{ex})^\gamma. \tag{9}$$

where the parameters are the same as in the listed-firms production function. The profit is also defined similarly to that of listed firms:

$$\pi^N(\Phi^{ex}) := \max_{k_i^T, k_i^I} \quad (k_i^T)^\alpha (k_i^I)^\theta (\Phi^{ex})^\gamma - rk_i^T - pk_i^I. \tag{10}$$

2.2.3 Market choice and the disclosure transparency

Firms choose between public and private markets to maximize value P, and if listing, select transparency level q. We use a submarket-choice formulation inspired by directed search (Menzio and Shi, 2010), with frictionless matching within each funding submarket. The resulting transparency dispersion among ex-ante identical firms is analogous to the price and wage dispersion in Burdett and Judd (1983) and Burdett and Mortensen (1998): firms are indifferent among the choices in the equilibrium support. In contrast, the non-listed market is served by only a mass $1/\nu_N$ of financiers, so non-listed firms face thinner funding participation. The firm-level decision is summarized as follows:

$$\max \left\{ \max_{q \in [0, 1-\overline{q}]} P(q), P^N \right\}. \tag{11}$$

where $P(q)$ is the price of the firm operating in the listed market with the transparency level at q, and P^N is the price of a non-listed firm, all of which are endogenously determined at the funding markets.

2.3 Funding markets

In this section, we characterize the financial market in the model. The funding supply is determined by the specialized financiers' local certainty-equivalent problems. The funding demand is determined by each firm's price (value) maximization. For the notational brevity, we omit the aggregate policy and allocation $(\overline{q}, \Phi^{ex})$ in the argument.

The following proposition specifies the symmetric local funding supplies for listed firms and non-listed firms. Because the mass assigned to each listed cell is normalized to one, the listed choice $x^*(q)$ is also the aggregate funding density for that submarket; aggregate private funding is x^{N*}/ν_N.

Proposition 1 (Funding supply)**.**
A specialized financier's optimal funding for listed firms with transparency q, $x^*(q)$, and the corresponding private-market funding, x^{N*}, are

$$x^*(q) = \frac{\overline{r}(q)}{\Lambda H(q;\overline{q})} = \frac{\pi(q)/P(q)}{\Lambda/(\xi + \psi(\overline{q}+q))}, \quad x^{N*} = \frac{\pi^N/P^N}{\Lambda/\xi}. \tag{12}$$

Proof. See Appendix G. ■

The optimal funding supply in each submarket increases in the expected profit and transparency and decreases in the price and the local risk aversion Λ.

In the funding submarket q for the listed firms, the price of a firm, $P(q)$, is determined at the level where funding supply in the number (measure) of firms $x^*(q)/P(q)$ meets funding demand in the number (measure) of firms $\mathcal{M}(q)$. Thus, the market-clearing condition is as follows:

$$\frac{x^*(q)}{P(q)} = \mathcal{M}(q). \tag{13}$$

Using Equations (12) and (13), the price maximization problem can be translated into an ex-ante profit maximization form as in the right-hand side formulation of the following line:

$$\max_{q\in[0,1-\overline{q}]} P(q) \iff \max_{q\in[0,1-\overline{q}]} \sqrt{\frac{\pi(q)}{\Lambda\frac{\mathcal{M}(q)}{\xi+\psi(\overline{q}+q)}}} \iff \max_{q\in[0,1-\overline{q}]} \pi(q)\phi^L(q) \tag{14}$$

where $\phi^L(q) := (\xi + \psi(\overline{q}+q))/\mathcal{M}(q)$ is defined as net funding intensity. The solution to this problem characterizes the funding demand in the listed market.

The price of a non-listed firm, P^N, is determined at the level where private financiers' funding supply meets private firms' demand. Because only a mass $1/\nu_N$ of financiers serves

the private market, aggregate private funding is x^{N*}/ν_N, and the market-clearing condition in the number (measure) of firms is as follows:

$$\frac{1}{\nu_N}\frac{x^{N*}}{P^N} = M_N, \tag{15}$$

where M_N is the number (measure) of non-listed firms and $\nu_N > 1$ is a structural parameter indexing limited financier participation in the private market. Each participating financier chooses x^{N*}; aggregate private funding purchases the claims of the M_N private firms, and no financing capacity is idle. A higher ν_N means thinner participation and a lower funding intensity per private firm. Consistent with the listed market setup, we define the funding intensity of the non-listed market as $\phi^N := \xi/(\nu_N M_N)$.

2.4 Summary of a firm's problem

Before production, each firm chooses whether to list (extensive margin) and, if listing, selects transparency level $q \in [0, 1-\overline{q}]$ (intensive margin). Non-listed firms avoid disclosure-induced rent exposure but face thinner financier participation: only a mass $1/\nu_N < 1$ of specialized financiers serves the private market, which lowers private funding intensity. Listed firms access the fully served public submarkets but expose competitively sensitive information, reducing appropriable rents while creating knowledge spillovers. A firm's problem is summarized as follows:

$$\begin{aligned}
&[\textit{Entry decision}] \quad \max\{J^L(\mathcal{M}), J^N(M_N)\},\\
&[\textit{Listed firm}] \quad J^L(\mathcal{M}) := \max_{q\in[0,1-\overline{q}]} \max_{k_T,k_I} \left(k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta}(\Phi^{ex})^{\gamma} - rk_T - pk_I\right)\phi^L(q)\\
&\qquad\qquad \text{s.t. } \phi^L(q) = \frac{\xi + \psi(\overline{q}+q)}{\mathcal{M}(q)},\\
&[\textit{Non-listed firm}] \quad J^N(M_N) := \max_{k_T,k_I} \left(k_T^{\alpha}(k_I)^{\theta}(\Phi^{ex})^{\gamma} - rk_T - pk_I\right)\phi^N\\
&\qquad\qquad \text{s.t. } \phi^N := \xi/(\nu_N M_N).
\end{aligned}$$

2.5 Equilibrium

We define an equilibrium where the economy is given total intangible capital reserve K^I (fixed aggregate intangible supply). This equilibrium endogenously determines the unit cost of R&D. The cost increases if all the other firms increase their spending in R&D, so developing new intangible (knowledge) is harder if more firms seek new knowledge. The rental rate for the tangible capital r is exogenously given.

Definition 1. *A collection of functions* $(x^*, x^{N*}, k_T, k_I, k_T^N, k_I^N, q, \mathcal{M}, M_N, p, P, P^N, \Phi^{ex})$ *is an equilibrium given* K^I *if*

1. $\left(x^*, x^{N*}\right)$ *solves the specialized financiers' local certainty-equivalent problems.*
2. $\left(k_T, k_I, q, k_T^N, k_I^N\right)$ *solves the listed and non-listed firms' problem.*
3. *The measure of listed firms choosing a transparency level* q *is consistent with* $\mathcal{M}(q)$ *for all* $q \in [0, 1-\overline{q}]$.
4. *The measure of non-listed firms* M_N *satisfies*
$$\int_0^{1-\overline{q}} \mathcal{M}(q)dq + M_N = 1.$$
5. *The unit cost of R&D of intangible capital* p *is determined by the following equation:*
$$K^I = \int_0^1 k_{I,i}di.$$
6. *Aggregate shared knowledge satisfies*
$$\Phi^{ex} = \int_0^1 1_{\{i \in Listed\}} \times k_{I,i}(\overline{q} + q_i)di.$$
7. *The funding markets are cleared:*
$$\frac{x^*(q)}{P(q)} = \mathcal{M}(q) \quad for\ \forall q \quad and \quad \frac{1}{\nu_N}\frac{x^{N*}}{P^N} = M_N.$$
8. *Indifference in the extensive-margin decision:*
$$P(q) = P^N,\ for\ \forall q \in [0, 1-\overline{q}).$$

The aggregate intangible capital market condition and the aggregation of the shared knowledge can be re-written with the equilibrium distribution $\mathcal{M}$:

$$K^I = \int_0^{1-\overline{q}} k_I(q, \mathcal{M})\mathcal{M}(q)dq + k_I^N M_N, \tag{16}$$

$$\Phi^{ex} = \int_0^{1-\overline{q}} k_I(q, \mathcal{M})(\overline{q} + q)\mathcal{M}(q)dq. \tag{17}$$

As specified by the indifference condition, the equilibrium of interest is the non-degenerate equilibrium where all the ex-ante homogeneous firms become indifferent over the choices of submarkets. In condition 8, we require the support of the equilibrium distribution $\mathcal{M}$ to be the full range of q, the closure of $[0, 1-\overline{q})$; the indifference condition is imposed on the half-open interval because retained rents, and hence the equilibrium density, vanish at the right endpoint.[10] In the following section, we analytically characterize the equilibrium allocations along with conditional uniqueness of the firm distribution.

Equilibrium determinacy Our equilibrium requires that the firm distribution $\mathcal{M}$ takes the full support $[0, 1-\overline{q}]$, which pins down a unique equilibrium distribution conditional on the aggregate objects (p, Φ^{ex}); the equilibrium is then unique up to the fixed point in these aggregates, whose uniqueness we do not establish (Corollary 2). This full-support condition parallels the dispersed price equilibrium in Burdett and Judd (1983): ex-ante identical firms randomize over the full range of transparency levels, and the resulting distribution equates the value of listing at every transparency level in the interior of the support. Without this condition, degenerate equilibria with mass points could arise, but these are not robust to small perturbations in the firm's problem and are therefore excluded as a standard equilibrium refinement.

3 Equilibrium analysis in closed form

In this section, we analytically characterize the equilibrium allocations and study the model predictions. A listed firm's problem is as follows:

$$\begin{aligned} \max_{q} \quad & \overbrace{\left[\max_{k_T, k_I} \left(k_T^{\alpha}(k_I(1-\overline{q}-q))^{\theta}(\Phi^{ex})^{\gamma} - rk_T - pk_I\right)\phi^L(q)\right]}^{\text{Interim problem}} \\ \text{s.t.} \quad & \phi^L(q) = \frac{\xi + \psi(\overline{q}+q)}{\mathcal{M}(q)}. \end{aligned} \tag{18}$$

where ϕ^L is the net funding intensity function from Equation (14). From the optimality conditions of the interim problem, we derive the analytical form of the intangible input demand as a function of transparency q and the regulation parameter $\overline{q}$ as follows:

[10] Without this equilibrium condition, the model does not rule out the possibility of multiple equilibria with distributions taking singleton supports.

Proposition 2. *(Intangibles and the transparency)*
The input demand of intangible capital k_I is as follows:

$$k_I(q, \mathcal{M}; \overline{q}) = \left(\left(\frac{\alpha(\Phi^{ex})^{\gamma}}{r} \right)^{\frac{1}{1-\alpha-\theta}} \left(\frac{r\theta}{p\alpha} \right)^{\frac{1-\alpha}{1-\alpha-\theta}} \right) (1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}. \tag{19}$$

Proof.
See Appendix G. ■

Greater disclosure-induced exposure dissipates a firm's appropriable rents by enabling competitors to respond. It does not destroy the intangible stock: rather, a higher normalized exposure index $q + \overline{q}$ lowers the private service flow $k_I(1 - q - \overline{q})$ generated by a given stock. Therefore, an increase in voluntary transparency leads to a decrease in intangible capital demand. This result predicts that the local financiers' forecast error (variance) about a firm's return and the firm's intangible demand are positively correlated in the cross-section of firms. Corollary 1 states this model prediction. Section 4 examines a related association in the data.

Corollary 1. *(Intangibles and the forecast error)*
Given the prices and the externality, the cross-sectional correlation between the local financiers' forecast errors and the firm-level intangible capital demand is positive.

Proof.
The proof is immediate from Proposition 2, given that the forecast error (variance) of a local financier is $H(q; \overline{q}) = 1/(\xi + (\overline{q} + q)\psi)$. ■

From the indifference condition with respect to the transparency q, we derive the following condition:[11]

$$\mathcal{M}(q) = \pi(q) \frac{\xi + \psi(\overline{q} + q)}{\pi^N \phi^N} \tag{21}$$

[11] Equation (14) combined with the equilibrium condition 8 yields the indifference condition. Alternatively, it can be expressed as follows:

$$\frac{\phi^{L'}(q)}{\phi^L(q)} = \frac{\theta}{1 - \alpha - \theta} \left(\frac{1}{1 - \overline{q} - q} \right). \tag{20}$$

This ordinary differential equation (ODE) determines ϕ^L up to a multiplicative constant; the constant and M_N are pinned down jointly by the extensive-margin indifference condition and the total mass condition:

$$\int_0^{1-\overline{q}} \mathcal{M}(q) dq + M_N = 1.$$

The proof in Appendix G proceeds via this ODE representation.

Solving the equation, we characterize the transparency distribution $\mathcal{M}$ in the analytic form as in the following proposition:

Proposition 3. *(Transparency distribution)*
The unnormalized probability density function $\mathcal{M}$ of transparency q has the following analytic form:

$$\mathcal{M}(q) = (\xi + \psi(\overline{q} + q))\,(1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}\,\frac{1}{\phi^N} \tag{22}$$

where $\phi^N = \frac{\xi}{\nu_N M_N}$.

Proof.
See Appendix G. ■

The endogenous firm distribution $\mathcal{M}$ is composed of multiplicatively separable components, each of which has a direct economic interpretation:[12]

$$\mathcal{M}(q) = \underbrace{(\xi + \psi(\overline{q} + q))}_{\text{local funding supply}}\ \underbrace{(1 - \overline{q} - q)^{\frac{\theta}{1-\alpha-\theta}}}_{\text{funding demand}}\ \underbrace{\frac{1}{\phi^N}}_{\text{eq. normalizer}}. \tag{23}$$

The first component captures the specialized financiers' greater local funding when transparency lowers the forecast-error variance (funding supply channel). This channel generates an incentive for a firm to choose high q. The second part captures a firm's incentive to reveal less information (funding demand channel): greater exposure dissipates the disclosing firm's appropriable rents as competitors respond. The third term is the equilibrium object that normalizes the measure of listed firms. The balance between the first two channels shapes the indifference, leading to the endogenous non-degenerate firm distribution $\mathcal{M}$ in the equilibrium, as in Burdett and Judd (1983).

In the funding demand channel, the transparency terms interact with the intangible share θ. If the intangible share is higher, the negative effect on the transparency density becomes greater. This is because the disclosure becomes costlier when the intangible becomes a more important input factor in the production due to the competition. In Section 5, we show how the intangible share and the disclosure policy interplay to shape the macroeconomic equilibrium allocations.

The following corollary establishes that, for the given support of the transparency and conditional on the aggregate objects, the equilibrium distribution is unique; the equilibrium

[12]It is worth noting that the endogenous distribution is independent of the productivity level or the externality. Although we abstract from the firm-level productivity heterogeneity, our equilibrium and the key mechanism are unaffected by such heterogeneity.

is then unique up to the fixed point in those aggregates.

Corollary 2. *(Conditional uniqueness of the firm distribution)*
Given the full support $[0, 1-\overline{q}]$ *and conditional on the aggregate objects* (p, Φ^{ex})*, the equilibrium firm distribution* $\mathcal{M}$ *is unique; the equilibrium is then unique up to the fixed point in these aggregates. Uniqueness of that aggregate fixed point is not established.*

Proof.
The proof is immediate from the fact that the firm's profit optimality condition and the equilibrium indifference condition imply the unique distribution in Proposition 3. ■

Moreover, we show that the probability density function $\mathcal{M}(q)$ belongs to a variant of a well-known distribution class, Beta distribution, which eases the equilibrium analysis even further. The following corollary shows that $\mathcal{M}(q)$ follows a shifted and scaled transformation of a truncated beta distribution.

Corollary 3. *(Truncated normalized Beta distribution)*
The gross transparency, $y := q + \overline{q}$*, is a shifted and scaled transformation of a random variable* Y *that follows a truncated* $Beta(B+1, 2)$ *distribution:*

$$q + \overline{q} = 1 - (1 + \xi/\psi) \times Y \tag{24}$$

where Y *has the* $Beta(B+1, 2)$ *density truncated to* $[0, \frac{1-\overline{q}}{1+\xi/\psi}]$ *and renormalized to integrate to one,* $B := \theta/(1-\alpha-\theta)$*, and the unnormalized firm measure* $\mathcal{M}$ *equals* $(1-M_N)$ *times the density of* q *induced by this distribution through the transformation* $q = 1 - \overline{q} - (1+\xi/\psi)Y$*.*

Proof.
See Appendix G. ■

Using the definition of the net funding intensity $\phi^N = \frac{\xi}{\nu_N M_N}$, total mass condition in the equilibrium can be written as

$$\frac{\nu_N}{\xi} M_N \psi \int_0^{1-\overline{q}} \left(\frac{\xi}{\psi} + (\overline{q} + q) \right) (1 - \overline{q} - q)^B dq = 1 - M_N. \tag{25}$$

Equation (25) captures the relationships between the structural parameters and the total measure of non-listed firms, M_N. After rearranging the terms, we obtain the analytic form of the measure of non-listed firms as stated in Proposition 4

Proposition 4 (Non-listed firms' measure)**.**
In equilibrium, the measure of non-listed firms M_N is

$$M_N = \frac{1}{1 + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B+2}\mathcal{B}(B+1,2)F\left(\frac{1-\overline{q}}{1+\xi/\psi}; B+1, 2\right)}, \tag{26}$$

where $\mathcal{B}$ is the beta function, F is the cumulative distribution function of beta distribution, and $B := \theta/(1-\alpha-\theta)$.[13]

Proof.
See Appendix G. ■

Notably, the analytic form of the non-listed firms' measure and the distribution of listed firms do not include either the price of the intangible or the externality. The intuition behind this result is that changes in price and externality uniformly shift the operating profits of listed and non-listed firms, so they do not cross-sectionally affect the decision of how to finance their operating activities. This separation mimics the block-recursive nature of the dynamic equilibrium under the directed search (Menzio and Shi, 2010). The distribution also does not depend on local financiers' risk-aversion parameter Λ, because it scales all funding cells symmetrically. M_N determines the funding intensity of private firm ϕ^N. Then, from Proposition 3, the firm distribution $\mathcal{M}$ is also fully characterized independent of the intangible price and the externality as follows:

$$\mathcal{M}(q) = \frac{\nu_N(\xi + \psi(\overline{q}+q))\,(1-\overline{q}-q)^{B(\theta)}}{\xi\left(1 + \psi\frac{\nu_N}{\xi}(1+\frac{\xi}{\psi})^{B(\theta)+2}\mathcal{B}(B(\theta)+1,2)F\left(\frac{1-\overline{q}}{1+\xi/\psi}; B(\theta)+1, 2\right)\right)}. \tag{27}$$

3.1 Analytical comparative statics with equilibrium feedback

How do changes in intangible intensity and disclosure regulation affect the equilibrium number of listed firms? Our analytical framework permits closed-form comparative statics across the equilibrium, accounting for feedback in both funding supply and demand that would require numerical methods in most models. The following proposition establishes

[13] The beta function is defined as follows:

$$\mathcal{B}(a,b) := \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)} = \int_0^1 x^{a-1}(1-x)^{b-1}dx.$$

monotone relationships between the measure of non-listed firms M_N and the structural parameters.

Proposition 5 (Drivers of listing decisions)**.**
For $\overline{q} \in (0,1)$ and $\theta, \psi, \nu_N, \xi > 0$, the equilibrium measure of non-listed firms M_N strictly

(i) increases in $\overline{q}$ (mandated transparency),

(ii) increases in θ (intangible share),

(iii) decreases in ψ (transparency's information content),

(iv) decreases in ν_N (limited private-market participation), and

(v) increases in ξ (baseline information level).

Proof.
See Appendix G. ■

Results (i) and (ii) formalize the paper's core mechanisms. Stricter mandated disclosure (higher $\overline{q}$) increases disclosure-induced rent exposure, driving firms to private markets. Similarly, greater intangible intensity (higher θ) amplifies this appropriability cost, resulting in a greater measure of non-listed firms. Result (iii) shows that when disclosure becomes more informative (higher ψ), more firms choose to list since the local funding benefit of transparency increases. Result (iv) is mechanical: worse private-market access makes listing relatively more attractive. Result (v) reflects that higher baseline information ξ reduces the relative information advantage of listed over private firms, shifting the equilibrium toward private markets. These sharp monotone predictions arise because the firm's problem exhibits increasing differences in listing incentives (Topkis, 1978; Milgrom and Shannon, 1994) that are preserved in the equilibrium (Acemoglu and Jensen, 2013).

In the following section, we analytically characterize welfare, transparency, and productivity, which are the relevant scoreboards for the policy design. While the comparative statics for the measure of (non-)listed firms are sharply characterized, their implications for these scoreboards involve competing forces and cannot be signed analytically. We therefore turn to quantitative analysis in Section 5, where we discipline the model with data to evaluate these trade-offs.

3.2 Welfare, transparency and aggregate productivity

In this section, we analytically characterize key macroeconomic measures to evaluate the equilibrium. Based on these measures, Section 5 quantitatively assesses the macroeconomic

implications of the rising intangibles and the disclosure policy. In the U.S., the SEC's objective is stated clearly in its mission: "The mission of the SEC is to protect investors, maintain fair, orderly, and efficient markets, and facilitate capital formation."[14] In line with the view of the SEC, we assess the effect of the disclosure regulation based on the investors' welfare and aggregate productivity.

First, we define the welfare measure. Our investor-welfare criterion is the equal-weighted, money-metric sum of local certainty equivalents, net of policy-invariant outside wealth. Each listed submarket is served by a unit mass of financiers, while a mass $1/\nu_N$ serves the private market, so the private certainty equivalent enters with weight $1/\nu_N$:[15]

$$\begin{aligned} W^F(\overline{q}) \equiv Objective_{welfare} &= \int_0^{1-\overline{q}} \left[x(\widetilde{q}) \frac{\pi(\widetilde{q})}{P(\widetilde{q})} - \frac{\Lambda}{2} x(\widetilde{q})^2 \frac{1}{\xi + \psi(\overline{q} + \widetilde{q})} \right] d\widetilde{q} \\ &\quad + \frac{1}{\nu_N} \left[x^N \frac{\pi^N}{P^N} - \frac{\Lambda}{2} (x^N)^2 \frac{1}{\xi} \right] \\ &= \frac{1}{2} \int_0^{1-\overline{q}} \mathcal{M}(\widetilde{q}) \pi(\widetilde{q}) d\widetilde{q} + \frac{1}{2} M_N \pi^N. \end{aligned} \tag{28}$$

The last equality uses each financier's first-order condition and the two funding-market clearing conditions. It is worth noting that W^F is not the expected utility of a representative investor. It adds the certainty-equivalent cost borne by each segmented financier before independent local payoff shocks wash out in the aggregate. Therefore, the measure is resource-consistent, as aggregate expected financier income equals the operating profits generated by listed and private firms.

The second measure is the aggregate transparency defined as

$$\mathcal{T} := \int_0^{1-\overline{q}} (q + \overline{q}) \mathcal{M}(q; \theta) dq. \tag{29}$$

A stricter mandate affects aggregate transparency through two margins. It raises the exposure of each listed firm but shrinks the pool of listed firms. Quantitatively, the participation effect dominates over the empirically relevant range of $\overline{q}$ (Section 5).

Transparency affects welfare through two channels. First, it lowers financiers' forecast-error variance, which raises funding and makes listing more attractive. Because this gain is capitalized into prices, it affects welfare only through the allocation of firms across markets. Second, disclosed knowledge spills over to other firms and raises their productivity, which is

[14] See "Our Goals," SEC website https://www.sec.gov/our-goals.

[15] The detailed derivation is available in Appendix F.

our last measure:

$$Objective_{productivity} = (\Phi^{ex})^{\gamma} = \left(\int_0^{1-\overline{q}} (\overline{q} + q) k_I(q) \mathcal{M}(q) dq \right)^{\gamma}. \tag{30}$$

The productivity measure is equivalent to the externality effect, which is a function of the total shared knowledge. From the regulator's perspective, there is a productivity trade-off in increasing the strictness of the disclosure requirement. For higher $\overline{q}$, each listed firm exposes a larger share of its knowledge, while the pool of listed firms to share the information shrinks due to the firm-level extensive margin responses.

4 Empirical evidence

The mechanism suggests a negative association between firms' reliance on intangible capital and the information available to outside investors. We present the firm-level reduced-form evidence in this section using firm-level data on publicly listed U.S. firms.

The firm-level data is from Compustat covering the period from 1985 to 2016 to measure firm-level intangible capital stock and other firm characteristics. Our stock measure has three separately constructed components: internally generated knowledge capital based on R&D expenditure, internally generated organizational capital based on a fraction of SG&A expenditure, and acquired intangibles based on reported intangible assets and amortization. We report the details on the construction (the perpetual inventory method) in Appendix A and the R&D-only trends in Appendix B. We report results both for the combined stock and for the three components entered separately.

In our model, intangible capital refers specifically to knowledge whose property rights are not fully protected by legal institutions. Patents and other formally protected intellectual property are excluded and treated as tangible assets. The empirical measure cannot separate legally protected from unprotected knowledge within each component, although the split between internally generated and acquired components reported below is informative about composition. The mapping to the model remains imperfect: the estimated association may partly reflect the composition of measured intangibles, and the data do not establish that restricting the measure to unprotected knowledge would strengthen the coefficient.[16]

To get a measure of forecast errors, we use the I/B/E/S to obtain data on earnings surprises. The dataset collects quarterly estimates made by professional financial analysts on the future earnings of publicly traded companies. From there, we closely follow DellaVigna

[16] In the structural estimation (Section 5), the intangible share parameter θ is disciplined by intangible-related *expenditures* rather than by the capitalized stock, but that choice does not eliminate this measurement limitation.

and Pollet (2009) for the definition and calculation of earnings surprises. Specifically, earnings surprise $ES_{i,j,t}$ is defined as the difference between a firm's announced actual earnings per share $e_{i,t}$ and the earnings forecast per share $\epsilon_{i,j,t}$ made by an analyst for that firm, normalized by the price of a share $P_{i,t}$:

$$ES_{i,j,t} := \frac{e_{i,t} - \epsilon_{i,j,t}}{P_{i,t}}, \tag{31}$$

where t is the indicator of a quarter; i and j are firm and analyst indicators, respectively. Thus, the surprise is measured at the analyst-firm level. Because price-normalized surprises have extreme tails when share prices are very low, we winsorize $ES_{i,j,t}$ within each calendar year at the 1st and 99th percentiles before constructing the proxies. Winsorization is applied to the surprises themselves rather than to the inverted proxies, as near-zero errors would otherwise translate into extreme inverse values. The proxies are constructed at the firm-quarter level, and the estimation sample contains one observation per firm-quarter, so firms with greater analyst coverage are not implicitly overweighted.[17]

Transparency is not directly observable. We therefore construct two complementary analyst-based information proxies based on the dispersion and accuracy of analysts' earnings forecasts, consistent with our model setup. Each proxy is a joint outcome of disclosure, the predictability of fundamentals, and analysts' information and behavior.[18] In Appendix E, we also explore a different measure of disclosure that is not based on our model but uses content analysis of corporate filings, the S&P Composite Transparency and Disclosure Index.[19] We find a negative association with this alternative measure as well, driven by its financial transparency and disclosure subcategory (Appendix E), although its S&P 500 coverage is limited to one year, 2002. Our first proxy measures the extent of analysts' disagreement about firm performance. Specifically,

$$AnalystInfo_{i,t}^{Disp} := \frac{1}{var(ES_{i,j,t})} \tag{32}$$

which is the inverse of the variance of earnings surprises for a firm i in a given quarter

[17] One firm-quarter enters twice because a fiscal-year change assigns the same quarter-end to two fiscal years; dropping either copy leaves all estimates unchanged at the reported precision.

[18] This approach also finds ground in the accounting literature, which directly links analysts' forecast accuracy and agreement to firms' disclosure quality and readability of financial reports. Lang and Lundholm (1996) find that analysts' forecast agreement and accuracy are positively related to the levels of disclosure of the company; Lehavy et al. (2011) show that analysts' earnings forecasts are less accurate when firms issue less readable 10-Ks.

[19] In 2002, as part of an initiative to introduce new analytical services on information, Standard & Poor's produced a measure of corporate disclosure based on the analysis of annual reports, 10-Ks, and proxy statements. The measure is available for firms in the S&P 500 index for the year 2002 only.

t. The intuition behind this proxy is that more transparent firms have lower dispersion (disagreement) in the earnings surprise among the analysts. This proxy is a data counterpart to the inverse of the subjective uncertainty $H(q;\overline{q})$ in our model setup in Equations (1) and (3). This proxy is measured only for firms with at least two analysts' forecasts available. In our dataset, the average number of analysts covering a firm is three.

The second is the inverse of the median distance between firm earnings and analyst forecasts:

$$AnalystInfo_{i,t}^{Error} := \frac{1}{median(|ES_{i,j,t}|)} \tag{33}$$

This proxy is based on the hypothesis that more transparent firms have lower absolute earnings surprises, on average.[20]

We limit the sample to firms in Compustat from 1985 to 2016 for which information on earnings forecasts by analysts is available. We exclude all financial firms (two-digit NAICS 52). We further require firms to have strictly positive assets, sales, and employment; extreme surprises are handled by the within-year winsorization described above rather than by a separate cutoff. Firm-quarters whose median winsorized surprise is exactly zero, about 10 percent of constructed firm-quarters, are dropped because the log-inverse proxy is undefined for them. The resulting sample contains 107,623 firm-quarter observations for 6,026 firms, and 57,983 observations for the dispersion proxy, which requires at least two analysts. We report summary statistics of the sample and the sample construction in Appendix D.

We then estimate the relationship between the analyst-based information proxies and intangible capital using the following specification:

$$\log \mathrm{y}_{i,t} = \alpha_t + FEs + \beta \times \text{Intangible over total assets}_{i,t} + \gamma \times X_{i,t} + \varepsilon_{i,t} \tag{34}$$

where $y_{i,t}$ is either our first or second analyst-based information proxy, measured at the firm-quarter level. α_t are calendar year-quarter fixed effects and FEs are industry fixed effects in the main specification. Standard errors are two-way clustered by firm and calendar year-quarter. $X_{i,t}$ contains standard firm controls, including book-to-market ratio, liquid capital (cash, inventory, and receivables), the liabilities-to-assets ratio, sales and employment in logs, age, the trailing five-year standard deviation of log sales growth lagged one year (requiring at least three growth observations), and the number of analysts covering the firm. Intangible and liquid capital are scaled by total assets.[21] Appendix D reports robustness to

[20] The median is used instead of the average; this is to rule out the outlier's level effect on the average.

[21] We control for total liabilities over assets rather than total debt over assets: the total-debt item is sparsely populated before 1997 in our constructed panel, and requiring it would effectively restrict the sample to the post-1997 period. The liabilities-to-assets ratio is fully populated throughout the sample; Appendix D documents this choice.

industry × year fixed effects, firm fixed effects, and collapsing the quarterly outcomes to the firm-year level.

Table 1: Analyst-based information proxies and intangible capital

	Baseline		Components	
	$1/\text{var}(ES)$	$1/\text{med}\|ES\|$	$1/\text{var}(ES)$	$1/\text{med}\|ES\|$
Intangible/assets	-0.378***	-0.219***		
	(0.111)	(0.035)		
R&D knowledge/assets			-1.008***	-0.269***
			(0.145)	(0.046)
Organizational/assets			-0.715***	-0.596***
			(0.241)	(0.087)
Acquired intangibles/assets			1.509***	0.613***
			(0.232)	(0.102)
Observations	57,983	107,623	57,983	107,623
R^2	0.316	0.318	0.324	0.324

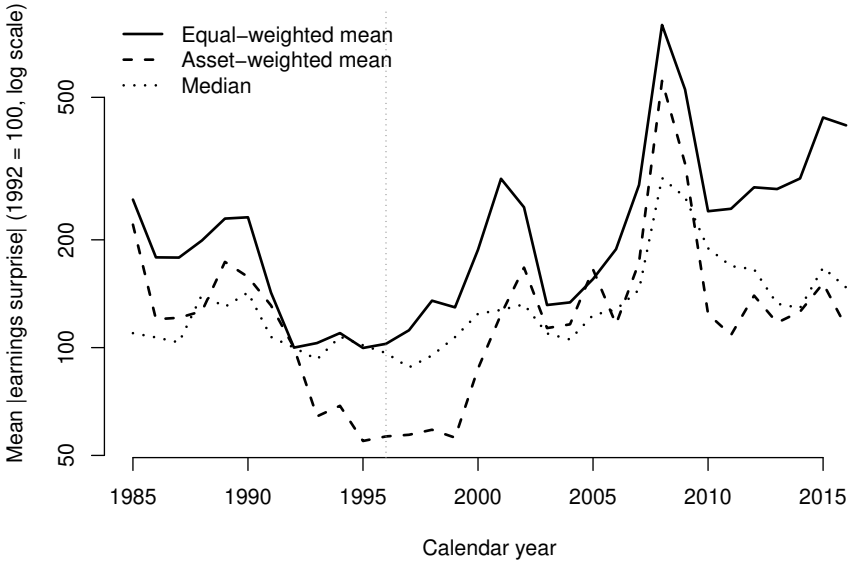
Notes: This table reports estimates from the firm-quarter panel over 1985–2016. The dependent variable is the log of the inverse variance of earnings surprises ($1/\text{var}(ES)$, available with at least two analysts) or the log of the inverse median absolute surprise ($1/\text{med}|ES|$). Both are analyst-based information proxies, not direct measures of chosen disclosure, and are built from surprises winsorized at the 1st and 99th percentiles within calendar year. The "Baseline" columns enter the combined intangible stock over assets; the "Components" columns enter R&D-based knowledge capital, SG&A-based organizational capital, and acquired intangibles jointly, each scaled by assets. All regressions include industry and calendar year-quarter fixed effects and the controls listed in the text. Standard errors (in parentheses) are two-way clustered by firm and year-quarter; *, **, and *** denote significance at the 10, 5, and 1 percent levels. Appendix D reports the components entered one at a time on the same sample, fixed-effect and aggregation robustness, and subsample results.

A key concern is that firms with high intangible intensity may also be inherently riskier, which may lead to a spurious correlation between intangibles and forecast errors. To address this issue, we control for firm-specific risk using the historical volatility of sales growth, defined as the standard deviation of log sales growth over the previous five fiscal years, lagged one year (Campbell et al., 2001; Irvine and Pontiff, 2009), and we report firm-fixed-effect estimates in Appendix D. Calendar year-quarter fixed effects absorb economy-wide changes in risk, including the common trend in firm-level volatility. Industry × year fixed effects, reported in Appendix D, also absorb industry-specific changes. Controlling for the number of analysts covering the firm captures part of the variation in analysts' total skill and effort.

Table 1 reports the results. In the baseline columns, the coefficient on the ratio of intangible capital to total assets is negative and statistically significant for both proxies. Quantitatively, an interquartile increase in the intangible capital over assets ratio is associated with a 19 percent lower inverse-dispersion proxy and an 11 percent lower inverse-error proxy. The results are robust to industry × year fixed effects, firm fixed effects, and collapsing the outcomes to the firm-year level (Appendix D).

The component columns refine this interpretation. The negative association is concentrated in internally generated intangible capital (both R&D-based knowledge capital and organizational capital), while acquired intangibles display the opposite, positive association: firms holding intangible assets that were acquired in market transactions and are recognized on the balance sheet are, if anything, easier to forecast. In interquartile terms, the joint estimates for the internally generated components range from 4 to 16 percent lower information proxies, against an 11 to 27 percent higher proxy for acquired intangibles. The three components are close to uncorrelated across firms, and entering them one at a time on the same sample yields nearly identical estimates (Appendix D). This contrast is consistent with the model's emphasis on unprotected, internally generated knowledge as the locus of disclosure-sensitive proprietary information.

Figure 2: Aggregate forecast error under alternative weighting



Notes: This figure plots the mean absolute earnings surprise across firms by calendar year, indexed to 1992 = 100 and shown on a logarithmic scale so that the persistent movement is visible alongside the 2008 spike. Surprises are price-normalized and winsorized at the 1st and 99th percentiles within calendar year before aggregation. The equal-weighted and asset-weighted lines weight firms equally and by total assets, respectively; the median line reports the cross-firm median. Financial firms (two-digit NAICS 52) are excluded. Data come from I/B/E/S and Compustat.

Two boundaries of the cross-sectional evidence should be noted. First, the association is concentrated among smaller and shorter-lived firms: it disappears in the cohort of firms observed in both the 1992–1996 and 2012–2016 windows, in the subsample of firms observed for at least fifteen years, and under asset weighting (Appendix D). This pattern is consistent with proprietary-information and valuation frictions mattering most for firms near the public-market participation margin. Because our homogeneous-firm model does not itself

predict such heterogeneity, however, we treat it as an interpretive observation rather than as a validation of the model. Second, the aggregate deterioration in forecastability that motivates the paper is not a small-firm artifact. Figure 2 shows that the mean absolute surprise rises between 1992–1996 and 2012–2016 by a factor of 3.3 when firms are weighted equally, 1.9 when they are weighted by assets, and 1.5 at the cross-firm median.

5 Quantitative analysis

We discipline our model by tightly matching the firm-level moments in the model with the data counterpart. Then, we use the disciplined model to report equilibrium counterfactuals and diagnostic decompositions of the observed macroeconomic changes. We estimate our model using data from two distinct periods. The first period, spanning from 1992 to 1996, serves as our baseline, and the second period, from 2012 to 2016, is used for the post-change period.[22]

5.1 Estimation

In this section, we elaborate on how we fit the firm-level data into the model. We assign values to the following parameters:

$$\{\theta, \overline{q}, \xi, \psi, \nu_N\}, \tag{35}$$

where θ is the intangible input share in production; $\overline{q}$ is a normalized index of regulatory exposure; ξ is the baseline precision of the information outside financiers have about both listed and non-listed firms; ψ is the disclosure's marginal contribution to financiers' information precision about listed firms; and ν_N indexes financier participation in the private market: a mass $1/\nu_N$ of specialized financiers serves it. We estimate θ, $\overline{q}$, and ψ by moment matching. We calibrate ξ to an out-of-model benchmark and normalize ν_N to an external financing-access rate, as detailed below. The five parameters are distinct; in particular, a change in ψ is not causally attributed to a change in θ.

Throughout this section, welfare denotes the measure defined in Section 3.2. It is not a comprehensive social-welfare measure.

[22] We end our sample in 2016 for two reasons. First, the post-2016 period features compositional shifts in IPO activity, including the SPAC wave of 2020-2021 and pandemic-related disruptions, that introduce measurement challenges for our transparency proxies. Second, our static model is designed to capture long-run equilibria, and the 2012-2016 window provides a stable post-adjustment period roughly 15-20 years after the structural break in listings began. Extending the sample through 2019 does not materially affect our baseline estimates.

The target moments and simulated moments are reported in Table 2. One of the important data moments we target is the portion of listed firms among the entire firms. Doidge et al. (2017) show that a significant portion of the U.S. listing gap is explained by the mergers and acquisitions (M&As) of listed firms. As our model does not have this channel, we adjust the data moment by residualizing the M&A effect. Starting from 1975, we sequentially updated the exit rate, which is the number of delisting firms minus M&As, over the number of listed firms, plus new entries and minus M&As. The total drop before the adjustment in listed firms was about 52%, but after accounting for M&As, the adjusted drop is 31%. This means that the adjusted fraction of listed firms went from 11.1% in the baseline period to 7.6% in the post-change period. Using the adjusted fraction of listed firms out of the total number of firms, we discipline the parameter $\overline{q}$, conditional on the other target moments.[23] The parameter should be read as a normalized regulatory-exposure index: a higher $\overline{q}$ raises required exposure and lowers the effective rent-retention factor $(1-\overline{q}-q)^{\theta}$. It is not the literal percentage of physical or knowledge stock disclosed, destroyed, or lost. Because $\overline{q}$ is identified from the listing margin, it should be read as a normalized index of regulatory exposure rather than as a measured disclosure rule.

In our model, the realized return faced by outside financiers in listed submarket q follows a normal distribution:

$$\widetilde{r}(q) \sim N\left(\overline{r}(q), \frac{1}{\xi + \psi(\overline{q}+q)}\right). \tag{36}$$

Analysts' forecast dispersion is a natural data counterpart to this return dispersion. Specifically, earnings surprise is defined as:

$$ES(q) := \overline{r}(q) - \widetilde{r}(q) \sim N\left(0, \frac{1}{\xi + \psi(\overline{q}+q)}\right). \tag{37}$$

Hence, in our analysis, we identify ψ using the average dispersion of the earnings surprise for each firm. These moments do not measure disclosure directly: they also reflect fundamental earnings uncertainty, analyst coverage, and heterogeneous beliefs. We assume that opacity in non-listed firms is comparable to that of the top opaque firms among the listed, which allows us to calibrate ξ out of the model. In particular, we calibrate ξ so that the zero-disclosure dispersion $\sqrt{H(0;0)} = 1/\sqrt{\xi}$ equals the 99th percentile of the firm-quarter dispersion distribution among listed firms (19.80 and 84.81 percent in the two periods). Within the fitted model, no listed firm attains this dispersion: the least transparent listed firm has dispersion $\sqrt{H(0;\overline{q})} = 1/\sqrt{\xi+\psi\overline{q}} < 1/\sqrt{\xi}$, so the tail benchmark serves as a calibration anchor rather than a moment the model reproduces. Appendix H documents

[23] We only consider firms with more than 100 employees following Doidge et al. (2017).

the identification analysis behind this treatment.[24] θ is identified from the intangible-expenditure-to-sales ratio. Lastly, we normalize ν_N so that the participating mass $1/\nu_N$ matches the 28.66% share of small businesses using non-depository financial institutions, based on the 1993 Survey of Small Business Finances (Cole and Wolken, 1995). It reports that 28.66% of small businesses used financial services from non-depository institutions, including finance companies, brokerage firms, and leasing companies. This access rate anchors the scale of private-market financier participation; it is not a firm-level match probability within the model. For the post-change period, we use a 4.1 percentage-point increase in normalized private-market financier participation, based on the estimate in Ewens and Farre-Mensa (2020).[25]

We use the method of simulated moments to estimate $(\theta, \overline{q}, \psi)$. The weight matrix is the identity matrix. Because the estimated block is exactly identified, the estimator attains zero distance at the solution and the weight matrix does not affect the point estimates (Newey and McFadden, 1994). The remaining two parameters are set outside the moment-matching block: ν_N is normalized to the observed financing-access rate, and ξ is calibrated to the zero-disclosure benchmark described above.

Table 3 reports the fitted parameters (estimated, calibrated, and normalized) along with the fixed parameters. In the post-change period, the regulatory-exposure index $\overline{q}$ increased from 0.982 to 0.995, consistent with the intended direction of post-1996 SEC reforms. These are normalized index values rather than literal disclosure percentages. The share of intangible input θ increased by approximately 84% (from 0.029 to 0.054), reflecting the significant rise in the importance of intangible input documented in the aggregate data.

The marginal information coefficient ψ declined from 38.5 to 11.4, meaning each unit of disclosure provides substantially less information precision to financiers in the post-change period. The target moments do not reveal the source of this decline, so we do not attribute it causally to θ. The baseline precision component ξ declined from 25.5 to 1.4, indicating a substantial increase in common residual information uncertainty that affects listed and non-

[24] The fitted moments in Table 2 are computed under the original construction of the estimation data: analyst-level surprises are aggregated to a firm-quarter root-mean-square surprise, winsorized at the 0.1st and 99.9th percentiles within each estimation window, and the calibration benchmark is the 99th percentile of that distribution. Under the revised construction of Section 4, the corresponding moments differ in level but preserve the qualitative pattern, and the tail-based level of ξ is sensitive to the cutoff choice. Re-estimating $(\overline{q}, \psi)$ on the revised-construction moments reproduces the targets exactly and leaves the policy conclusions of Section 5.4 intact (the required-benefit wedge becomes 16.4 and 35.1 percent), but it drives the fitted indices to the upper boundary, where the voluntary-disclosure band degenerates; we retain the original construction for the baseline fit.

[25] They identified the propensity to raise capital from out-of-state investors around NSMIA (Table 1). According to their analysis, NSMIA has increased the probability that a late-stage financing round includes an out-of-state investor by 4.1 percentage points, relative to early-stage rounds.

Table 2: Fitted moments

Moments	Data	Model	Reference
Baseline ($1992 \sim 1996$)			
Fraction of listed after M&A adj. (%)	11.08	11.08	Compustat & BDS
(*cf. without M&A adj. (%)*)	(8.30)		
Intangible Exp./Sale (%)	2.906	2.906	Compustat
Average $sd(\widetilde{r})$ (%)	12.53	12.53	Compustat & I/B/E/S
Zero-disclosure benchmark sd (%, calibration)	19.80	–	Compustat & I/B/E/S
Private financing-access rate (%, normalization)	28.66	–	Cole and Wolken (1995)
Post-change ($2012 \sim 2016$)			
Fraction of listed after M&A adj. (%)	7.60	7.60	Compustat & BDS
(*cf. without M&A adj. (%)*)	(4.01)		
Intangible Exp./Sale (%)	5.356	5.356	Compustat
Average $sd(\widetilde{r})$ (%)	28.00	28.00	Compustat & I/B/E/S
Zero-disclosure benchmark sd (%, calibration)	84.81	–	Compustat & I/B/E/S
Private financing-access rate (%, normalization)	32.76	–	Ewens and Farre-Mensa (2020)

Notes: The sources of the data target moments are specified in the last column. We join the datasets of Compustat and I/B/E/S following DellaVigna and Pollet (2009).

listed firms symmetrically; this component is distinct from both disclosure and fundamental volatility. The parameter ν_N decreased from 3.489 to 3.053, indicating broader financier participation in the private market, reflecting the impact of the National Securities Markets Improvement Act of 1996 (Ewens and Farre-Mensa, 2020).

Besides the estimated parameters, we fix the following parameters before the estimation:

$$\{\alpha, \gamma, r, K^I\}. \tag{38}$$

Tangible capital share, α, is set to be $0.40 - \theta$ for both baseline and post-change periods.[26] The knowledge-spillover elasticity γ is calibrated to twice the private intangible share, $\gamma = 2\theta$: estimates of the ratio of social to private returns to R&D of roughly 2.6 to 4 (Bloom et al., 2013; Lucking et al., 2019) imply γ/θ between 1.6 and 3 under our production structure. This evidence motivates the calibration but does not directly identify the model's γ/θ; we set $\gamma/\theta = 2$, in the lower half of the range. The equilibrium firm distribution and all estimated parameters are invariant to γ, which affects the intangible price, factor demands, output,

[26]Because our model abstracts from labor input, we read α and θ as output elasticities, with labor held fixed at the firm level. Under this reading, θ equals the ratio of intangible expenditure to sales, $\alpha + \theta = 0.40$ is the capital share, and the residual $1 - \alpha - \theta = 0.60$ accrues to claimholders and includes the return to the fixed labor input.

Table 3: Fitted, calibrated, and normalized parameters

Param.	Description	Baseline ($1992 \sim 1996$)	Post-change ($2012 \sim 2016$)
Estimated, calibrated, and normalized parameters			
$\overline{q}$	Regulatory-exposure index	0.982	0.995
θ	Intangible share	0.029	0.054
ψ	Transparency's contribution to public info.	38.533	11.396
ν_N	Private-market participation index (normalized)	3.489	3.053
ξ	Zero-disclosure info. level (calibrated)	25.520	1.390
Fixed parameters			
α	Tangible capital share	$0.40 - \theta$	$0.40 - \theta$
γ	Knowledge-spillover elasticity (calibrated)	$2\theta = 0.058$	$2\theta = 0.107$
r	Capital rental cost	0.10	0.07
K^I	Total intangible supply	1	1

Notes: We first set the fixed parameters as reported at the bottom of the table. We then estimate $(\theta, \overline{q}, \psi)$ by the method of simulated moments based on the matched targets in Table 2, while ξ is calibrated to the zero-disclosure benchmark and ν_N is normalized to the external financing-access rate, as described in the text.

productivity, and welfare, but not the firm distribution or the estimated parameters.[27] We assume the baseline period's capital rental cost (including depreciation) is 0.10, and 0.07 for the post-change period, reflecting the lowered capital cost in the recent periods. The total intangible capital stock, K^I, is normalized to 1.

5.2 Decomposition analysis

In this section, we decompose the macroeconomic changes between periods into diagnostics for each structural parameter. Because the effect of a given parameter depends on the levels of all other parameters, the exercise is order-dependent: the magnitude of a parameter's impact differs depending on whether baseline or post-change parameters are used as the reference point. To address this issue, we compute each parameter's diagnostic using a symmetric two-step procedure. First, beginning from the set of the baseline parameters, we adjust only one parameter to its post-change value and record the resulting change in the relevant outcomes (the number of listed firms, transparency, productivity, and welfare).[28] Second, starting from the set of the post-change parameters, we revert that parameter to

[27] Cross-country evidence that intangible capital generates productivity spillovers (Corrado et al., 2017) supports a positive γ. Evidence of small aggregate externalities (Basu and Fernald, 1995) argues against the upper end of the range, and because knowledge also diffuses through channels other than disclosure, the ratio-based mapping, if anything, overstates γ. Both considerations favor the lower half of the range, over which Section 5.4 reports sensitivity.

[28] Welfare, productivity, and transparency are defined in Section 3.2.

its baseline value and compute the counterfactual change. The symmetric diagnostic for each parameter is then given by the average of these two effects. Therefore, this symmetric procedure is not an additive decomposition: interactions imply that the five rows need not sum to the total change.

Table 4: Symmetric one-at-a-time diagnostics for macroeconomic changes

		Contribution to the change (p.a.):			
Param.	Channel	#listed	transparency	productivity	welfare
	Total change (%)	-1.88	-1.85	-0.95	-1.08
$\overline{q}$	Regulatory exposure	-6.04	-6.00	-0.55	-0.68
θ	Rising intangible share	-1.03	-1.03	-0.84	-1.91
ψ	Harder to forecast public firms	-3.72	-3.72	-0.35	-0.39
ν_N	Private financier participation	-0.61	-0.61	-0.05	-0.06
ξ	Baseline information level	8.62	8.62	0.74	0.83

Notes: Numbers are reported in percentage per year. Each parameter row is the symmetric one-at-a-time diagnostic described in the text. The θ row is an intangible-production block: it moves $\alpha = 0.40 - \theta$ and the spillover elasticity $\gamma = 2\theta$ together with θ. The welfare column reports the welfare measure defined in Section 3.2.

Table 4 presents the diagnostic results in annualized percentage points.[29] The fraction of listed firms declined by 1.88 percent annually (log change) over the 20 years between sample periods. The other columns report the model-implied changes in transparency, productivity, and welfare, which exhibited annual declines of 1.85, 0.95, and 1.08 percent, respectively. The model shows that a decline in transparency is accompanied by a loss in productivity, as less knowledge is shared among firms. This model prediction is consistent with the slowdown in U.S. productivity growth and business dynamism (Gordon, 2017; Bloom et al., 2020; Akcigit and Ates, 2023).

Among the rows that lower listings and transparency, the $\overline{q}$ row is the largest: its shift alone is associated with an annual 6.04% drop in the number of listed firms and a 6.00% drop in transparency. The θ row, an intangible-production block that moves $\alpha = 0.40 - \theta$ and $\gamma = 2\theta$ together with θ, is associated with annual changes of -1.03% in listed firms, -1.03% in transparency, -0.84% in productivity, and -1.91% in welfare. The ψ row is associated with -3.72%, -3.72%, -0.35%, and -0.39% in the same outcomes. We keep the ψ row separate from the θ row and do not label it an intangible-caused effect. The ν_N row is small: -0.61%, -0.61%, -0.05%, and -0.06%.

On the contrary, the ξ row contributed positively to the changes: $+8.62\%$ in listed firms

[29] The two periods of comparison are 20 years apart from each other. So, we annualized the total change by a division of 20.

and transparency, +0.74% in productivity, and +0.83% in welfare. This is because the declined baseline precision makes financiers provide little funding to the non-listed market, which makes the firms tend to go listed.

Both the rising intangible share and the stricter regulation index lowered productivity in these diagnostics. Among the rows that lower productivity, the largest impact was from the intangible share in the production function, followed by the regulation index.

A 1.08% annual reduction in welfare accompanied the observed changes. Within the diagnostics, the θ row lowers this measure by 1.91% and the $\overline{q}$ row by a further 0.68%. Higher $\overline{q}$ gives the financiers of listed firms more precise information. This gain is capitalized into funding terms, however, so it does not raise welfare directly. At the fitted indices, which lie above the welfare optimum to be discussed in the next section, the losses dominate: fewer firms list, and less knowledge diffuses.

5.3 Optimal disclosure policy and divine coincidence

In this section, we use our model to analyze the optimal level of disclosure regulation for welfare maximization. The policy index $\overline{q}$ has two-sided effects on welfare. On the one hand, higher mandated exposure lowers firms' rent retention and moves firms into the private market, muting knowledge spillovers. On the other hand, it feeds disclosed knowledge into the shared pool and sharpens financiers' information about the listed firms.

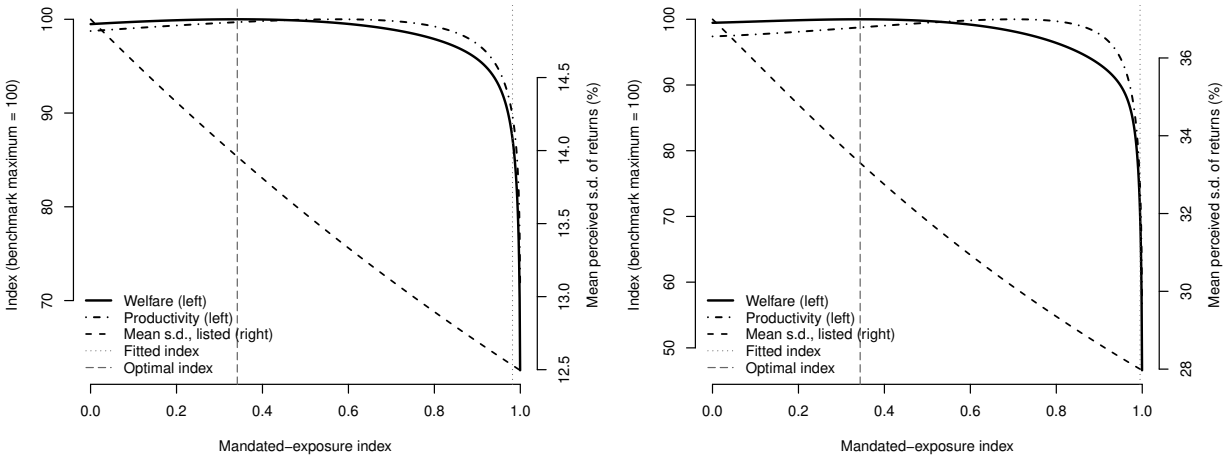
Figure 3 shows that welfare varies non-monotonically with regulatory stringency, exhibiting an inverted-U relationship reminiscent of a Laffer curve.[30] The optimum is at an index of about 0.34 in both periods, significantly below the fitted indices of 0.982 and 0.995. The hump is asymmetric. To the left of the optimum the surface is nearly flat: a zero mandate forgoes only about 0.5 percent of welfare. The flatness reflects an envelope property: financiers' information benefits are capitalized into funding volumes and prices, so the welfare value of a moderate mandate comes from knowledge diffusion rather than from information precision itself. To the right of the optimum the losses are an order of magnitude larger: the participation response dominates, and welfare declines steeply toward the fitted indices.[31] Depending on the contemporaneous policy level $\overline{q}$, moving $\overline{q}$ toward the welfare-optimal point can increase output and productivity as well, achieving a *divine coincidence* (Blanchard and Galí, 2007). Productivity is also hump-shaped in the index, peaking near 0.58 in the baseline period and 0.70 in the post-change period. At the fitted

[30] When tax rates are too low, revenue is limited by the rate; when too high, revenue is limited by the tax base. Similarly, insufficient mandated disclosure forgoes knowledge diffusion, while excessive disclosure drives too many firms private, shrinking the pool of disclosing firms.

[31] We report sensitivity results for the spillover calibration in Section 5.4.

indices, which lie above both peaks, the coincidence holds: decreasing $\overline{q}$ raises welfare and productivity together. The coincidence breaks down between the welfare optimum and the productivity peak, where a further decrease raises welfare but reduces productivity. The mean perceived standard deviation of listed returns falls as the index rises, so the trade-off at the fitted mandates is between welfare and investor information precision.

Figure 3: Welfare, productivity, and information along the regulatory-exposure index



(a) Baseline (1992–1996)

(b) Post-change period (2012–2016)

Notes: This figure plots welfare (solid line, left axis) and productivity (dash-dotted line, left axis), each normalized to 100 at its maximum, together with the mean perceived standard deviation of listed returns (dashed line, right axis), along the normalized regulatory-exposure index. The vertical dotted line marks the fitted index and the vertical long-dashed line marks the optimal index. Welfare and productivity are both hump-shaped, peaking in the interior before declining; information precision improves in the index.

Within the model, a moderate mandate is therefore desirable: knowledge diffusion supports an interior optimum. The observed mandates, however, lie significantly above the level this channel can justify. The model omits margins that motivate disclosure regulation in practice, including fraud deterrence, agency discipline, and information benefits to agents outside the funding market, and any welfare rationale for the observed mandates must rest on these excluded margins. The next subsection quantifies how large such benefits would have to be.

5.4 Heightened policy stakes in the opaque economy

In this section, we analyze how the observed changes affected the stakes of the disclosure policy. Welfare attains its maximum at an index of about 0.34 in both periods, significantly below the fitted indices of 0.982 and 0.995. The exercise is a quantitative illustration conditional on the calibrated spillover elasticity and on the maintained measurement model.

We quantify the distance as an *accounting gap*: the welfare gap between the fitted index and the optimum, holding each period's fitted parameter vector fixed. The gap measures the welfare cost of the mandate and, equivalently, the size of offsetting benefits outside this criterion that would be needed to rationalize the fitted index.

The gap is large in both periods and rises over time. Under the 1992–1996 parameters, the fitted index delivers 87.3 percent of benchmark welfare, so offsetting benefits worth about 13 percent of benchmark welfare would be needed to rationalize it. Under the 2012–2016 parameters, the fitted index delivers 70.5 percent of benchmark welfare, raising the required offset to about 29 percent. The wedge more than doubled between the two periods. The doubling reflects both a larger overshoot and a steeper welfare curve: as intangible capital became more central to production, the same deviation from the optimum generates larger welfare losses. A larger intangible share θ raises the rent-retention cost of exposure, and a lower ψ weakens the funding-side gains from additional mandated disclosure; the lower ξ works in the opposite direction, making the private alternative less attractive. The wedge rises with the calibrated spillover strength: for γ/θ between 1.6 and 3, it ranges from 9.8 to 19.8 percent in 1992–1996 and from 23.2 to 43.3 percent in 2012–2016, and even under the conservative reference $\gamma = \theta$ it is 5.3 and 13.0 percent. The location of the optimum is stable, between 0.24 and 0.48, across this range. Holding γ constant across periods isolates the role of the calibration. With γ fixed at its 1992–1996 value the wedge rises from 12.7 to 14.5 percent, and the post-change optimum moves to the lower boundary, so the interior optimum does not survive in the later period. With γ fixed at its 2012–2016 value the wedge rises from 24.4 to 29.5 percent and the optimum remains interior in both periods (0.47 and 0.34). The more-than-doubling of the headline gap therefore reflects the joint rise of θ and the spillover elasticity tied to it.

The information side of the ledger moves in the opposite direction. Moving from the optimal index to the fitted index lowers the mean perceived standard deviation of listed returns from 14.0 to 12.5 percent under the baseline and from 33.3 to 28.0 percent under the post-change parameters. A policymaker who values information precision beyond its capitalized funding effect faces this trade-off. The rising wedge measures the increase in the cost of this information objective between the two periods.

This analysis bears two caveats. First, the wedge is computed within a static model, holding fitted parameters fixed at each period's values; it compares long-run equilibria, not transitions. Second, the welfare measure omits distributional effects, other stakeholders, and implementation costs. Within this criterion, the cost of the observed mandate is large, it more than doubled as the economy became intangible-intensive, and rationalizing the mandate would require benefits of at least that size outside the criterion.

6 Concluding remarks

The disappearance of U.S. public firms and the declining transparency of those that remain signal a fundamental tension between investor protection and knowledge diffusion in an intangible-intensive economy. This paper develops an analytically tractable framework to characterize this tension and derive the optimal disclosure policy when firms face competing pressures: financiers demand transparency for informed investment decisions, while firms guard proprietary knowledge to maintain competitive advantage. Our analysis yields two central findings. First, the interaction between the rising intangible capital share and stricter disclosure requirements is associated with the observed decline in listed firms and market transparency between 1992-1996 and 2012-2016, imposing substantial productivity costs in the model. The estimated declines in baseline information quality and disclosure informativeness are consistent with a deeper transformation: intangible assets are inherently more difficult to value and more vulnerable to competitive spillovers than traditional capital. Second, welfare exhibits an inverted-U relationship with regulatory stringency: insufficient mandated disclosure forgoes knowledge diffusion, while excessive disclosure drives firms to opaque private markets. The optimal mandate lies at an exposure index of about 0.34 in both periods, and the observed disclosure environment lies significantly above it. The model therefore delivers an accounting gap: benefits outside the model's welfare criterion, such as fraud deterrence, agency discipline, or information for non-participants, would need to be worth about 13 percent of benchmark welfare to rationalize the fitted 1992-1996 mandate and about 29 percent for the 2012-2016 mandate, more than double. Critically, the welfare curve has steepened: as intangible capital becomes more central to production, precise policy calibration becomes increasingly consequential, and the participation response of firms is a first-order consideration in policy design.

Looking forward, as intangible capital continues its secular rise, welfare trade-offs intensify, heightening the stakes of disclosure policy. Our analytical framework provides a ground for understanding these trade-offs and characterizing optimal policy in an intangible-intensive economy.

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